

PAPER_469 — NGC 346 Nebula: MUGE UQFF Protostar Formation via Ug3 Collapse, Cluster Ugi Entanglement, and Blueshifted Quantum Waves

Whitepaper Series: Star-Magic UQFF Phase 2 — SMC Star-Forming Nebula
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Source: grok_share_dc707f5d3.txt (Doc 71/76 — NGC346UQFFModule, “MUGE NGC 346 Nebula”)
Classification: FIRST MUGE UQFF for NGC 346 SMC star-forming region; FIRST Ug3 gravitational collapse protostar term; FIRST UQFF blueshifted quantum wave ($v_{\text{rad}} = -10$ km/s) in quantum term; FIRST pseudo-monopole cluster entanglement via Ugi
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C++ Module: NGC346UQFFModule.h / NGC346UQFFModule.cpp

Abstract

NGC 346 is the most active star-forming region in the Small Magellanic Cloud (SMC), hosting $\sim 50,000$ young stellar objects within a 5 pc radius. This paper presents the MUGE UQFF gravitational model for NGC 346, incorporating protostar formation via the Ug3 gravitational collapse term, cluster entanglement via multi-body Ugi forces, blueshifted quantum waves ($v_{\text{rad}} = -10$ km/s, corresponding to infall toward the cluster center), and pseudo-monopole inter-cluster communication. The UQFF blueshifted quantum term is a **first application** of radial velocity Doppler coupling into the UQFF quantum wavefunction. Result: $g_{\text{NGC346}} \approx 1 \times 10^{-10} \text{ m/s}^2$ (collapse/wave dominant; Ugi entanglement advances framework).

2. Core Physics — PAPER_469

2.1 System Parameters

Parameter	Value	Notes
M (total)	$1000 M_{\odot}$ ($\sim 1.989 \times 10^{33} \text{ kg}$)	Cluster mass
r	5 pc ($\sim 1.543 \times 10^{17} \text{ m}$)	Cluster radius
SFR	$0.1 M_{\odot}/\text{yr}$	Active protostar formation rate
ρ_{gas}	$1 \times 10^{-20} \text{ kg/m}^3$	Gas density
v_{rad}	-10 km/s (-10^4 m/s)	Radial infall velocity (blueshift)
z	0.0006	SMC redshift (local)
M_{DM}	$\sim 0.85 \times M$	Dark matter fraction
B	$1 \times 10^{-5} \text{ T}$	Nebular magnetic field

2.2 Protostar Formation Gravitational Equation

$$g_{\text{NGC346}}(r, t) = \frac{GM_{\text{sf}}(t)}{r^2} (1 + H_z t) \left(1 - \frac{B}{B_{\text{crit}}} \right) (1 + F_{\text{env}}(t))$$

$$+U_{g3,\text{collapse}} + \sum_i U_{gi,\text{entangle}} + \frac{\Lambda c^2}{3} + U_i + g_{\text{quantum,blueshift}} + g_{\text{fluid}} + g_{\text{DM}}$$

2.3 Ug3 Gravitational Collapse Term (Protostar Formation)

The Ug3 term models the gravitational collapse driver for protostar formation:

$$U_{g3,\text{collapse}} = \frac{GM_{\text{proto}}}{r_{\text{Jeans}}^2}$$

Where $r_{\text{Jeans}} = \sqrt{15k_B T / (4\pi G \rho_{\text{gas}} m_H)}$ is the Jeans length. This is the **first explicit UQFF protostellar collapse term**, reducing the multibody protocloud to a Jeans-scale gravitational driver.

2.4 Ugi Cluster Entanglement

NGC 346's ~50,000 YSOs are modeled as quantum-entangled gravitational sources through the Ugi sum:

$$\sum_i U_{gi} = U_{g1} + U_{g2} + U_{g3,\text{collapse}} + U_{g4,\text{react}} + U_{\text{pseudo}}$$

Where U_{pseudo} = pseudo-monopole entanglement term:

$$U_{\text{pseudo}} = k_{\text{monopole}} \cdot \frac{N_{\text{YSO}}}{r^2}$$

This models the collective quantum gravitational communication between the ~50,000 YSOs as a pseudo-monopole network — the **first UQFF cluster entanglement term** for a distributed star-forming complex.

2.5 Blueshifted Quantum Wave Term ($v_{\text{rad}} < 0$)

The blueshift of the infalling gas ($v_{\text{rad}} = -10$ km/s) modifies the quantum wavefunction frequency:

$$k_{\text{blueshift}} = \frac{2\pi}{\lambda_{\text{dB}}} \cdot \left(1 + \frac{|v_{\text{rad}}|}{c}\right)$$

$$\psi_{\text{total}}(r, t) = A e^{-r^2/(2\sigma^2)} e^{i(k_{\text{blueshift}} r - \omega t)}$$

$$g_{\text{quantum,blueshift}} = \frac{\hbar}{\sqrt{\Delta x \cdot \Delta p}} \cdot |\psi|^2 \cdot \frac{2\pi}{t_{\text{Hubble}}}$$

The blueshift factor $(1 + |v_{\text{rad}}|/c)$ amplifies the de Broglie wavenumber, increasing quantum gravitational coupling during infall — **first Doppler-corrected UQFF quantum term**.

2.6 F_env: Collapse + Wave Pressure

$$F_{\text{collapse}} = \frac{M_{\text{proto}} G}{r_{\text{Jeans}}^2} \cdot \frac{\text{SFR}}{M_0}$$

$$F_{\text{wave}} = \rho_{\text{gas}} \cdot v_{\text{rad}}^2 / r$$

$$F_{\text{env}}(t) = F_{\text{collapse}} + F_{\text{wave}}$$

3. Equation Summary

$$g_{\text{NGC346}}(r, t) = \frac{GM_{\text{sf}}(t)}{r^2} (1 + H_z t) (1 - B/B_{\text{crit}}) (1 + F_{\text{collapse}} + F_{\text{wave}}) + U_{g3}^{\text{collapse}} + \sum_i U_{gi}^{\text{entangle}} + \dots$$

Computed Result: $g_{\text{NGC346}} \approx 1 \times 10^{-10} \text{ m/s}^2$ — Ug3 collapse and quantum wave dominant; Ugi cluster entanglement framework advances UQFF to distributed protostellar systems.

4. Physical Interpretation

- **Protostar collapse driver:** Ug3 Jeans collapse term is the dominant gravitational term at the protostellar scale ($r \sim 0.1 \text{ pc}$), overtaking the global g_{base} by $10\times$.
 - **Cluster entanglement:** The 50,000 YSOs of NGC 346 communicate gravitationally through pseudo-monopole states — the UQFF models this as a collective Ugi network operating simultaneously across the 5 pc cluster volume.
 - **Blueshifted infall:** The $v_{\text{rad}} = -10 \text{ km/s}$ Doppler blueshift increases quantum coupling during the active star-formation phase, accelerating collapse via enhanced $k_{\text{blueshift}}$.
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5. C++ Module Reference

Module: NGC346UQFFModule (root-level, Session 120 from grok_share_dc707f5d3.txt)

Key method: computeG(double t) — returns total g_{NGC346} in m/s^2

Unique feature: Blueshifted quantum wavefunction, pseudo-monopole entanglement Ugi

Integration point: MAIN_1_CoAnQi.cpp SMC star formation validation

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Nebular/Star-forming region luminosity $H\alpha$ + X-ray GR	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	L_X SFR observable	HST/ALMA/Chandra	Consistent order of magnitude
Schwarzschild limit κ vacuum rate vs X-ray variability	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	$r_s = 2GM/c^2$ (GR exact) Observed X-ray variability τ_{obs} (instrument monitoring)	PDG 2024 / GR HST/ALMA/Chandra	$\square$ UQFF respects GR horizon Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Nebular/Star-forming region through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST/ALMA/Chandra monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

QS=5 — Full UQFF integration: Ug3 collapse, Ugi cluster entanglement, blueshifted quantum wave, SMC NGC346 protostar formation. *Copyright — Daniel T. Murphy. Encoded Oct 10, 2025.*